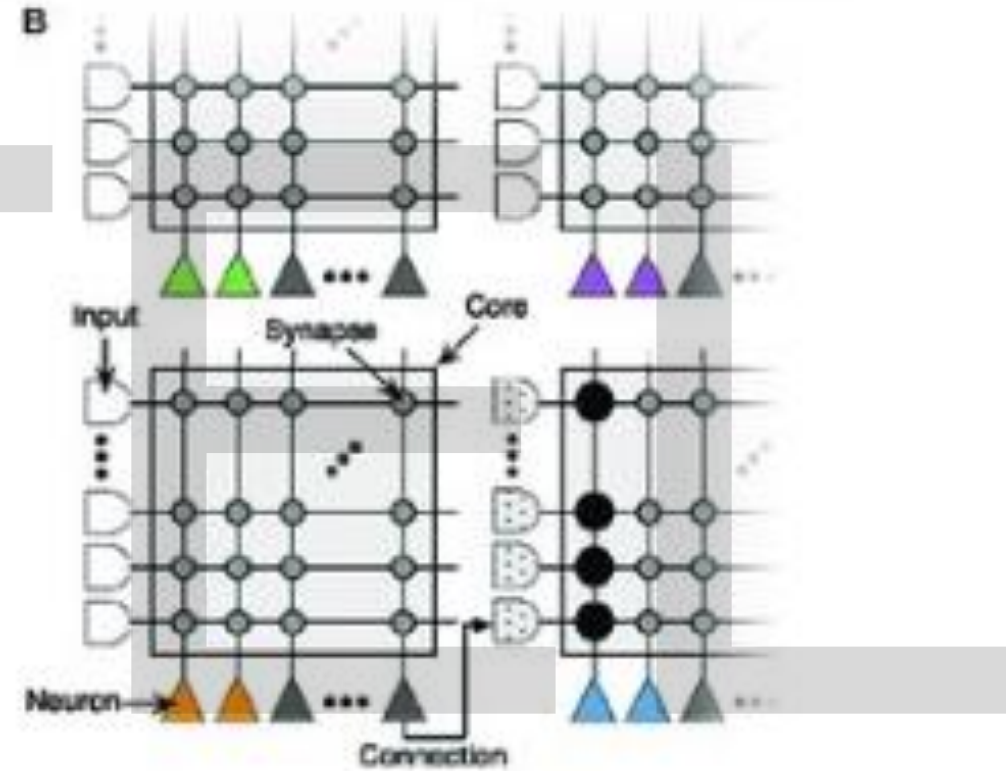


Spiking Architecture

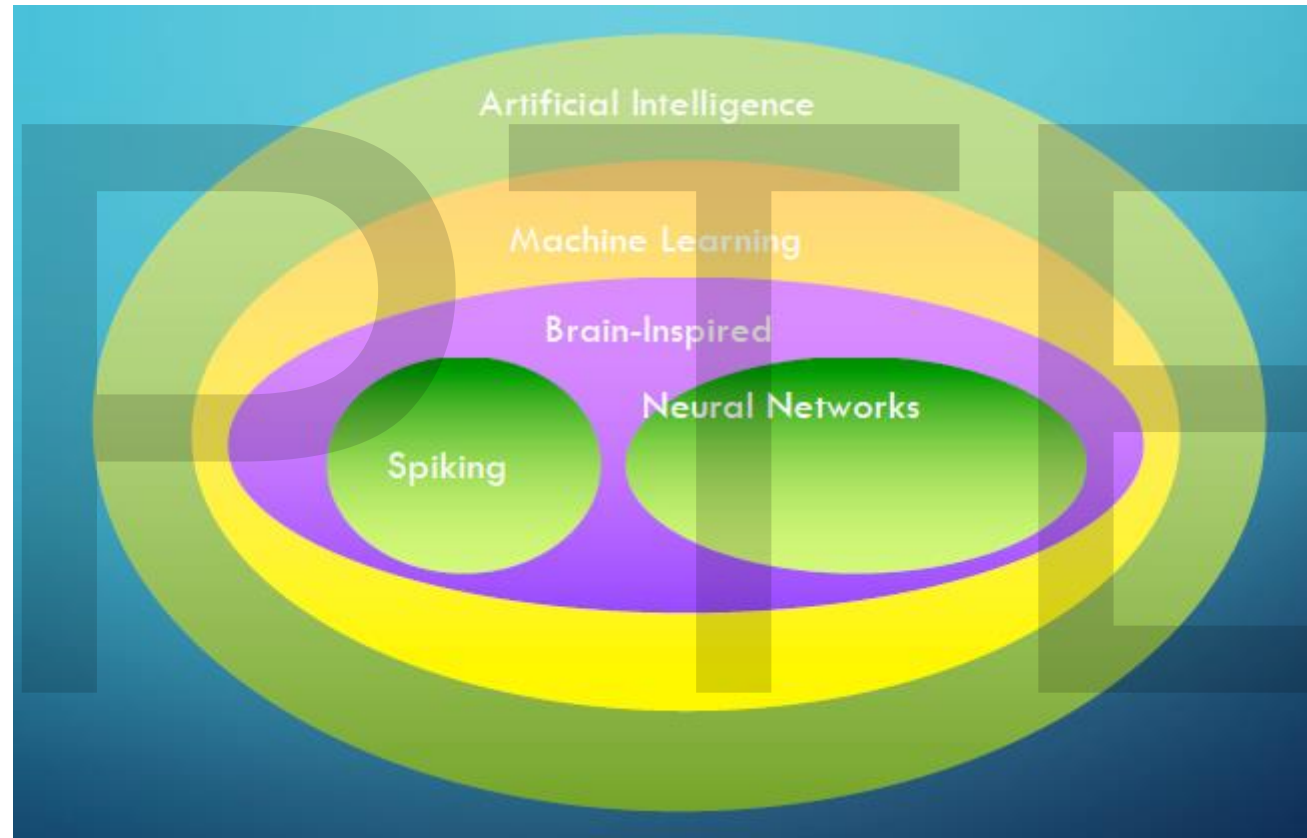
- Brain-inspired
- Integrate and fire
- Example: IBM True North
 - 1M neurons
 - 256M synapses
 - 5.4B transistors
 - 73 mW power consumption



TrueNorth Core Architecture

Neural Networks

N



L

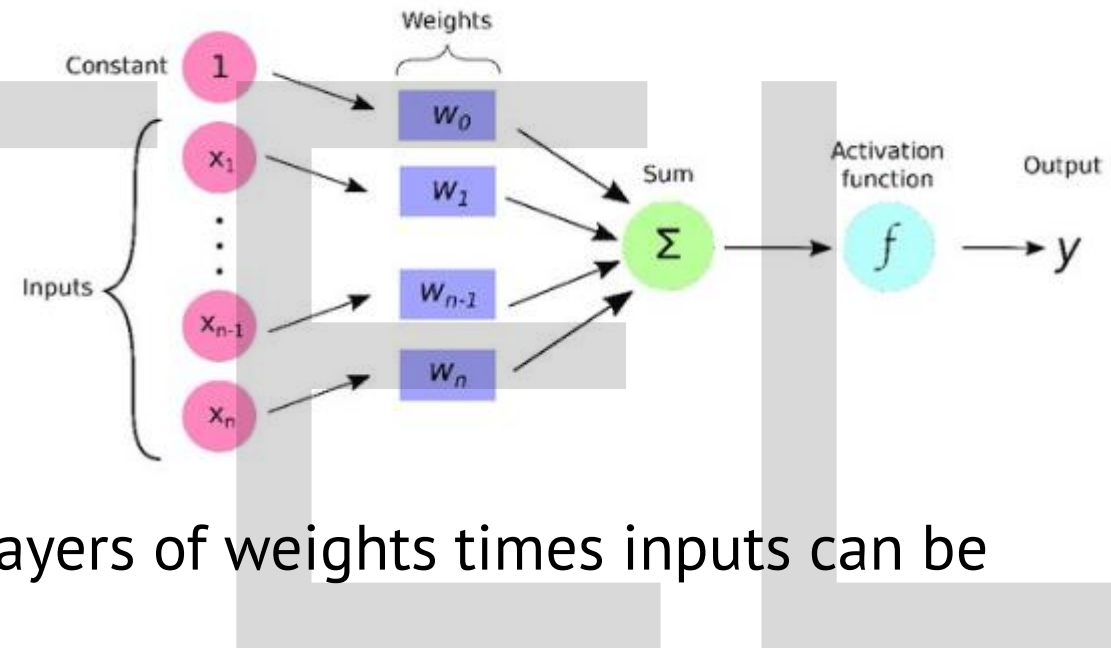
Perceptron Proposed by Rosenblatt (1957)

- **Perceptron is the building block of neural network**

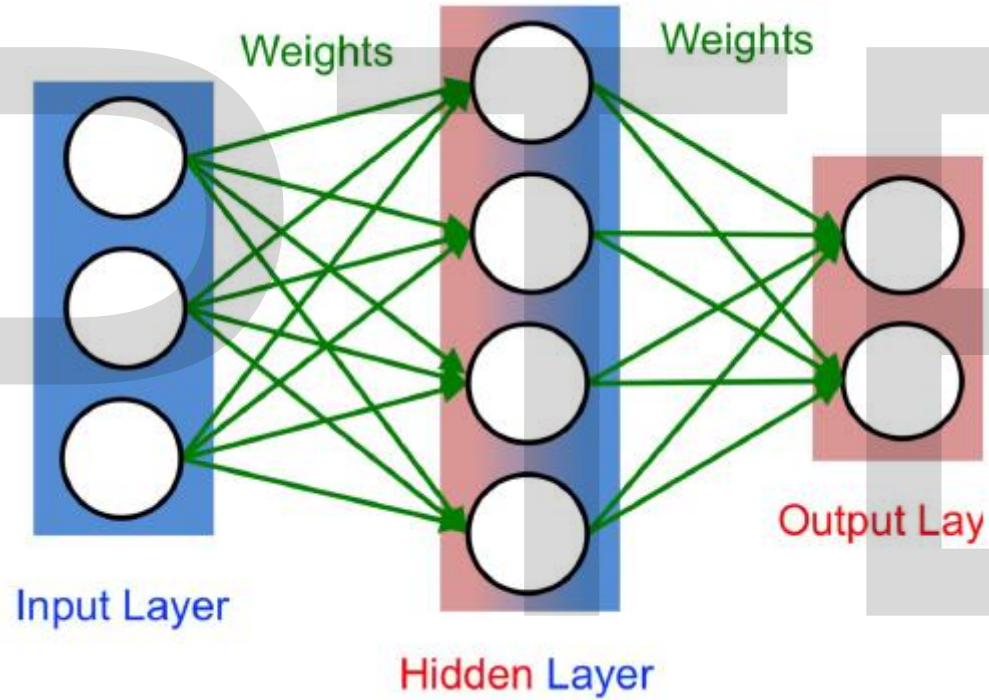
- Inputs multiplied by weights
- Added
- A non-linear activation function is applied
- Output is obtained

- **Role of non-linearity**

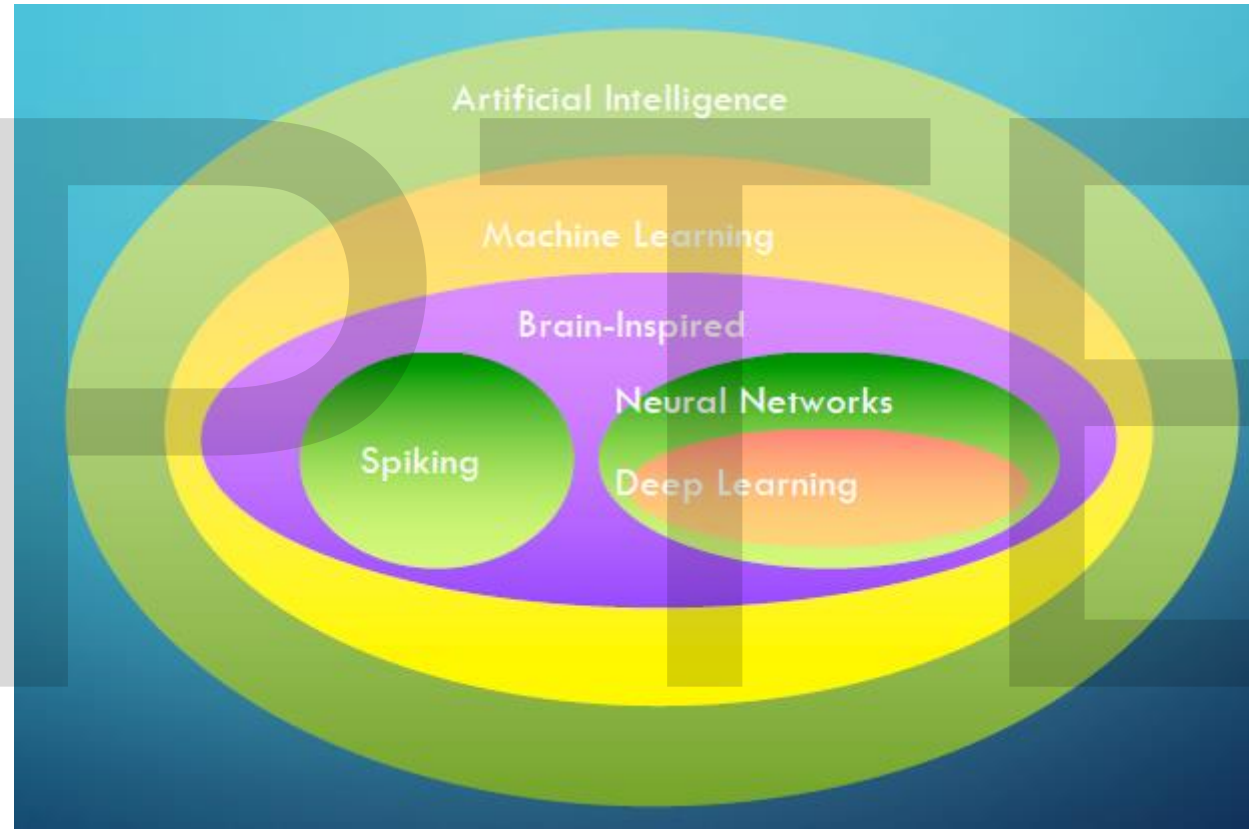
- If there is no non-linearity then multiple layers of weights times inputs can be reduced to single layer
- Take-home thinking!



Multi-layer Perceptron



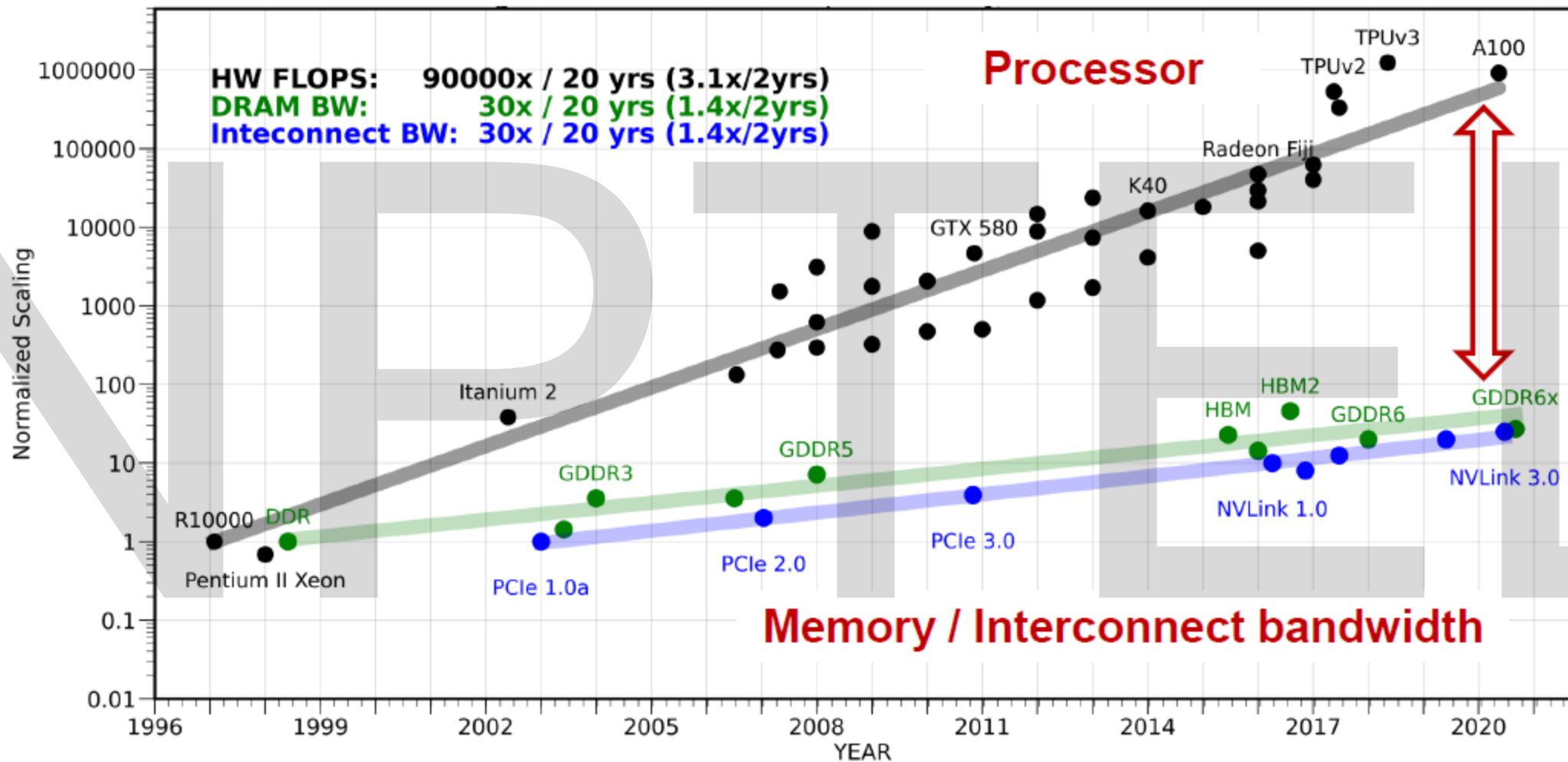
Deep Learning



Types of Learning

- **Supervised learning**
 - Learn a model M from labelled input x,y
- **Unsupervised learning**
 - Clustering, outlier detection
- **Generative learning**
- **Reinforcement learning**
 - Output $\rightarrow$ action, label $\rightarrow$ reward
-

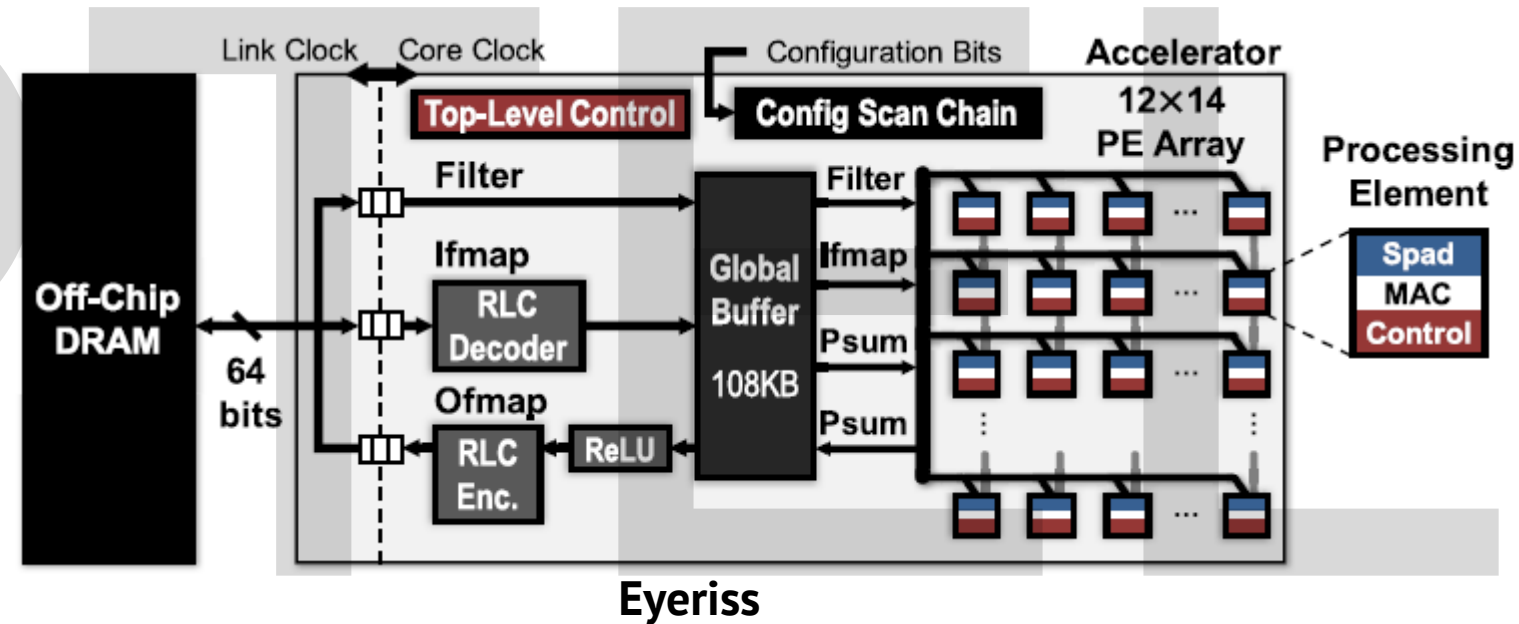
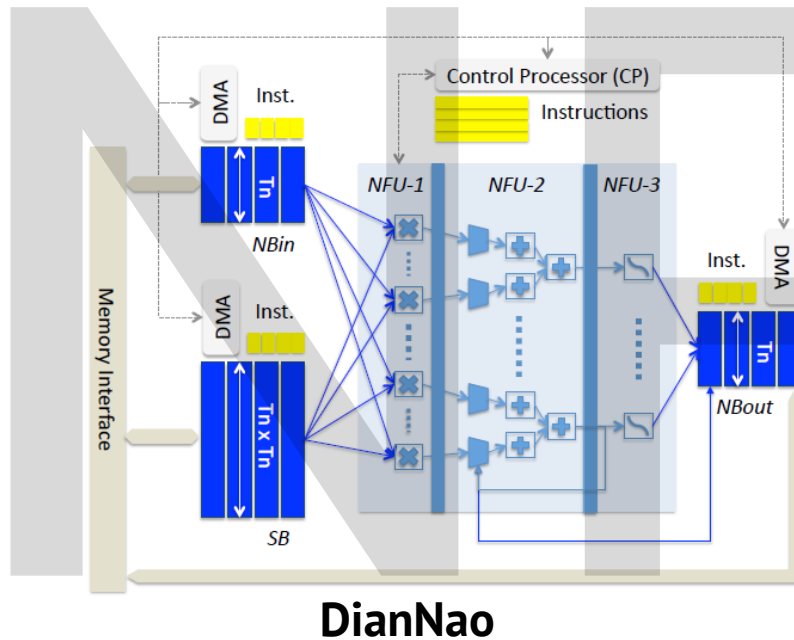
Challenges in Computing



Work by A. Gholami, 2020; Hynix; Chipworks; Created by Prof. Yu Cao, ASU.

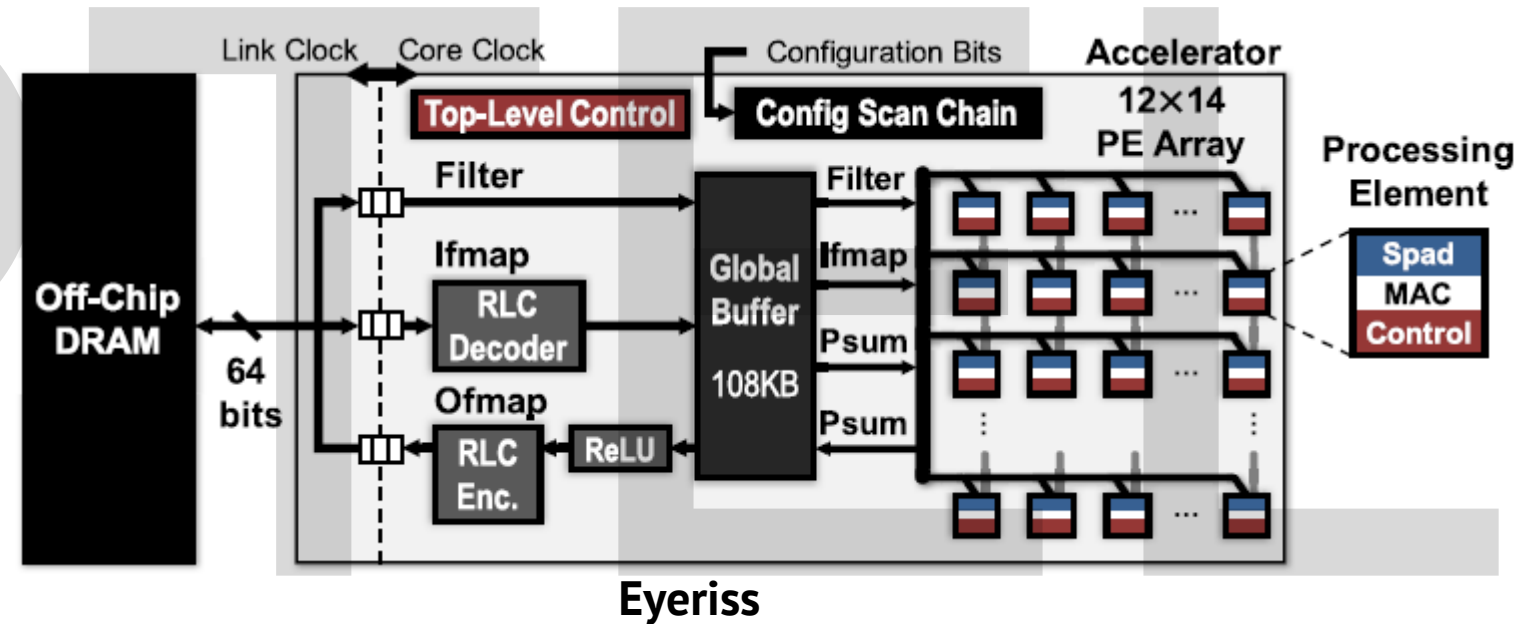
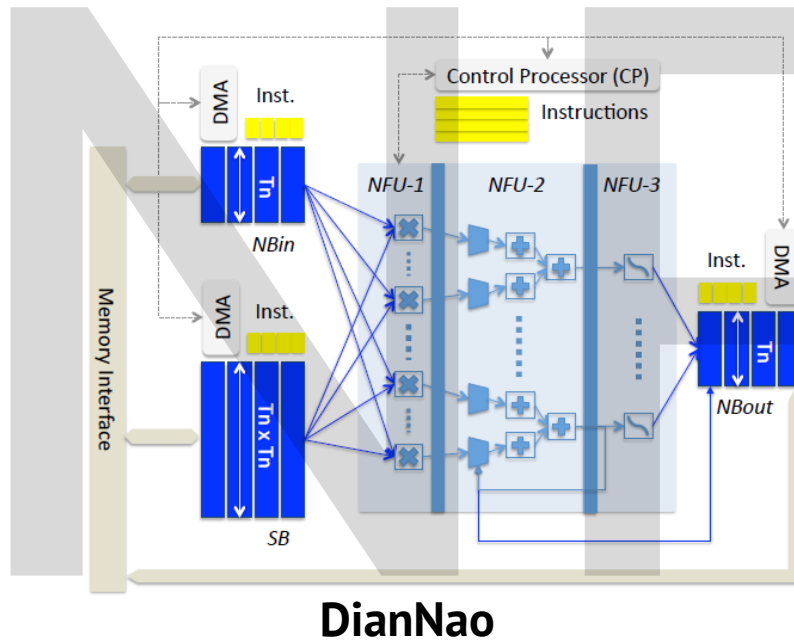
Accelerators

- Specialized processors to perform a specific task or group of task



Accelerators

- Specialized processors to perform a specific task or group of task



Still memory is a bottleneck

Figure 1 illustrates the architecture of the proposed memristor-based neuromorphic computing system. The top part shows a 4x4 array of memristor crossbars (XB) connected to an external IO interface chip (node). A detailed view of a single node shows its internal components: Input Register (IR), Output Register (OR), Max Pool Unit (MP), Shift and Add (S+A), Sigmoid Unit (σ), Sample and Hold (S+H), Digital to Analog Converter (DAC), and Analog to Digital Converter (ADC).

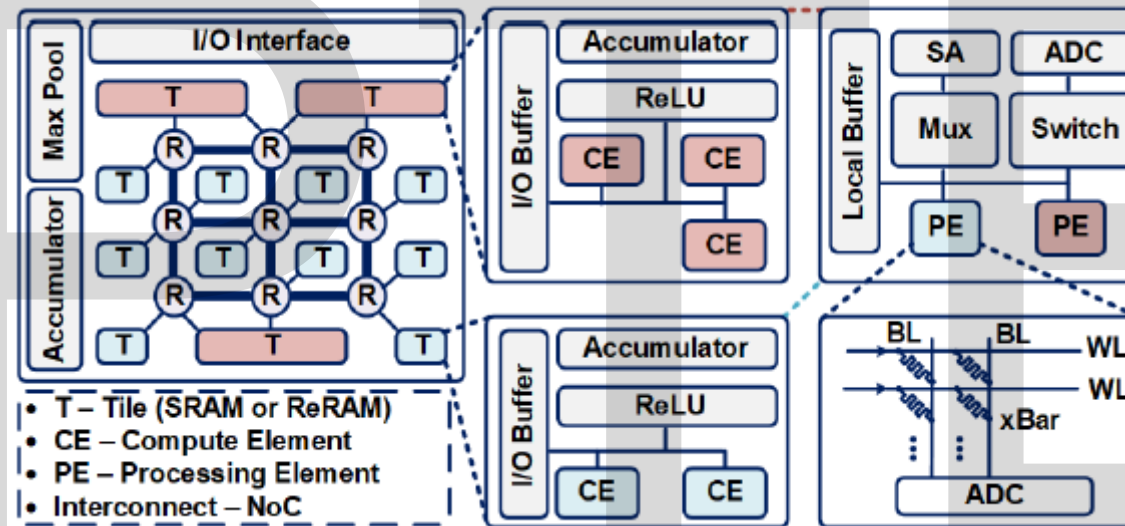


Figure 1 illustrates the architecture of the proposed memristor-based neuromorphic computing system. The top part shows a 4x4 array of memristor crossbars (XB) connected to an external IO interface chip (node). A detailed view of a single node shows its internal components: Input Register (IR), Output Register (OR), Max Pool Unit (MP), Shift and Add (S+A), Sigmoid Unit (σ), Sample and Hold (S+H), Digital to Analog Converter (DAC), and Analog to Digital Converter (ADC).



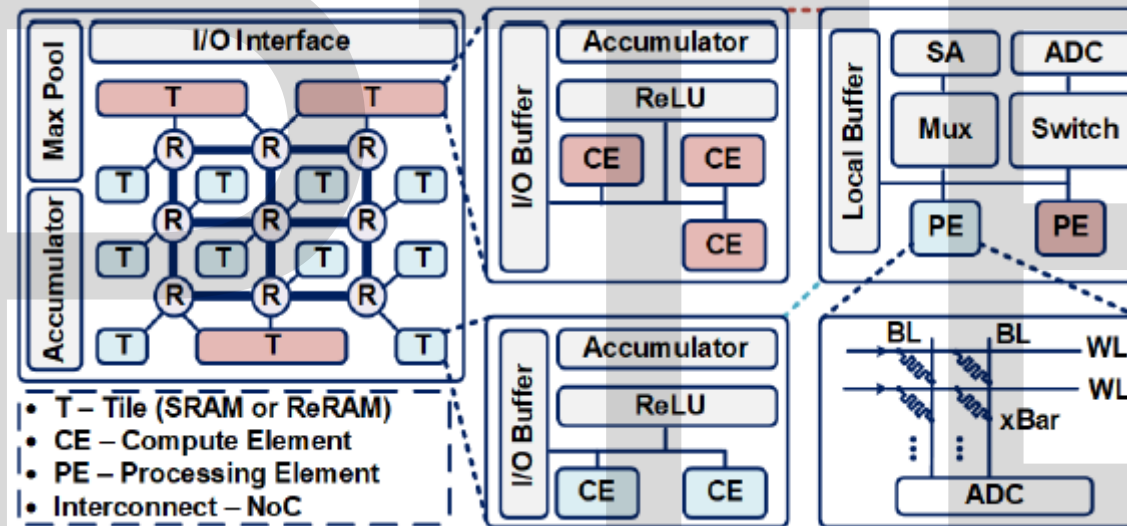
11

Communication-aware Accelerators



DnT'20

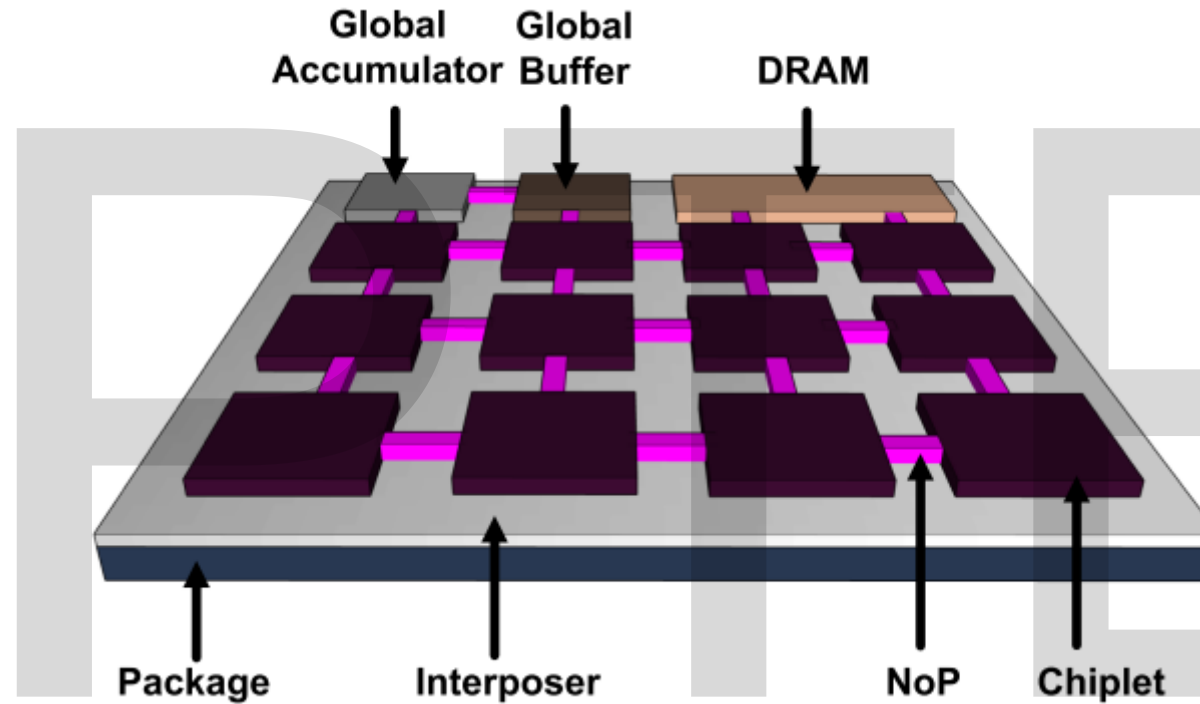
Communication-aware Accelerators



DnT'20

Area is a bottleneck

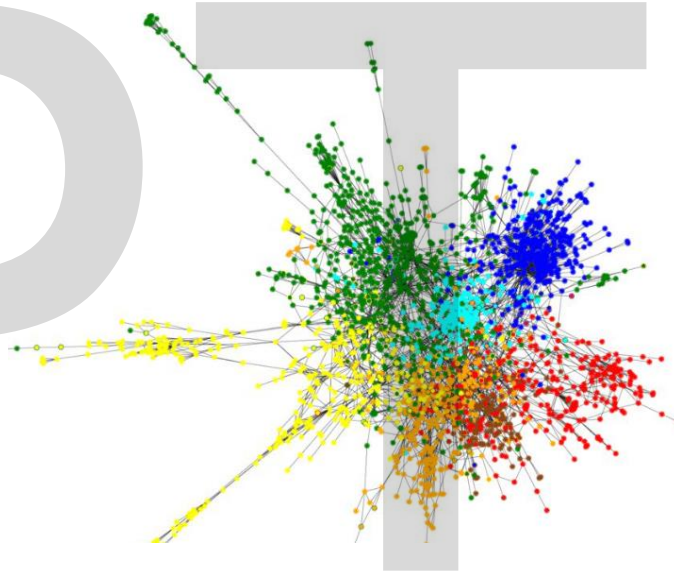
Fabrication Cost-aware Accelerator



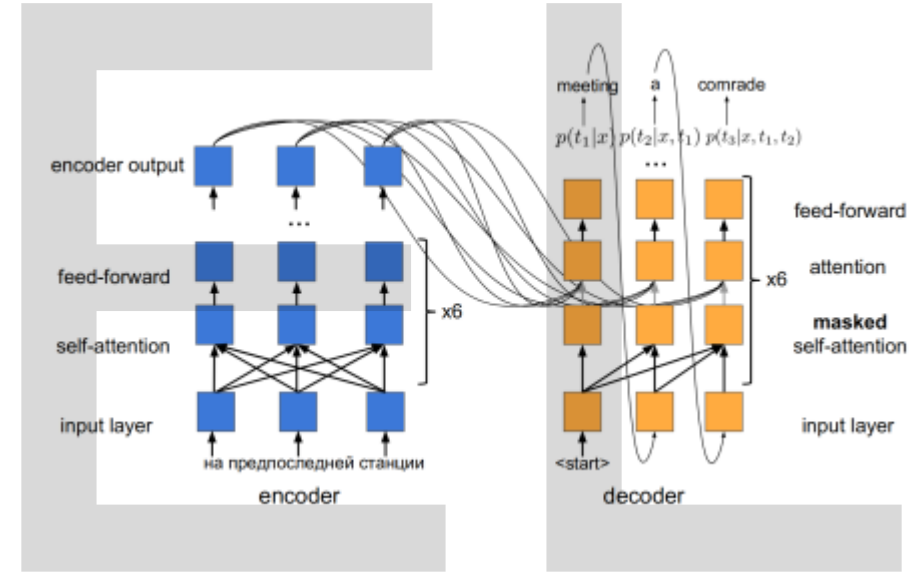
Story Does Not End Here

- **Complex algorithms**
 - Graphs
 - Transformer algorithms
- **Hardware challenges**
 - Power
 - Fabrication
 - Packaging

Example of Graph



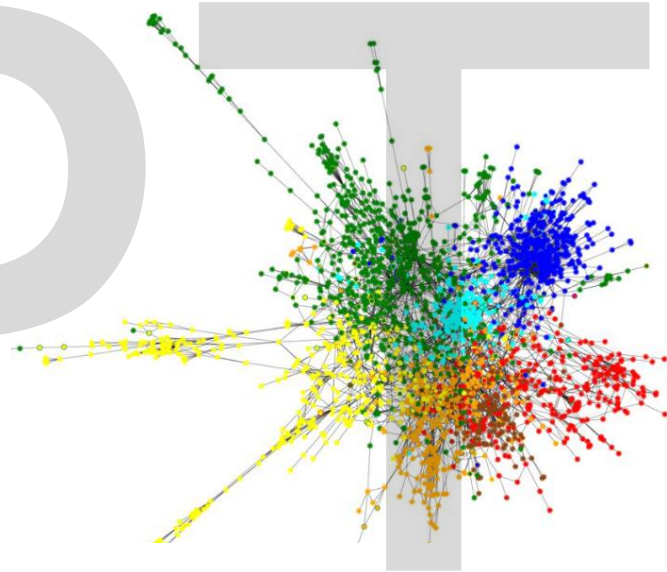
Example of Transformer Algorithm



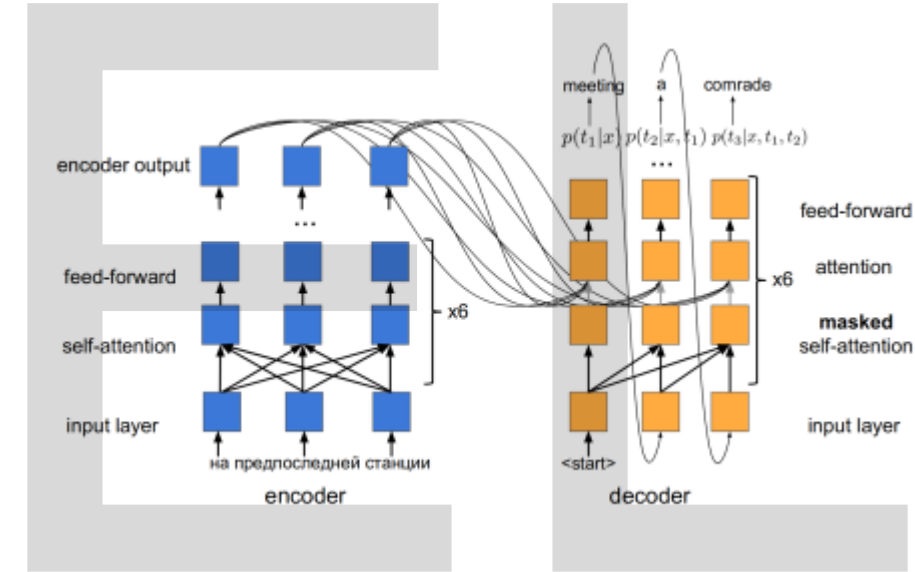
Story Does Not End Here

- **Complex algorithms**
 - Graphs
 - Transformer algorithms
- **Hardware challenges**
 - Power
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Example of Graph



Example of Transformer Algorithm



Will End up Deep Diving on These Topics